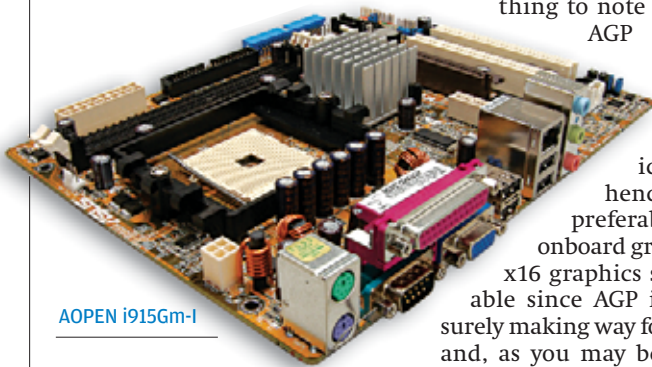




AOPEN i915Gm-I

erboard and it locks firmly too, ensuring that the heat-sink and the CPU remain in full contact. In the Socket 478 interface, the heat-sink and fan assembly mounting is a comparatively cumbersome thing to do.

Of the chipsets available for this platform, no points for guessing this one—Intel dominated across the range of the motherboards. The Intel chipsets ranged from the ageing Intel 845 to the newer Intel 915GV. We had to underclock the CPU so we could test the Intel 845 motherboards, since they did not support the 400 MHz FSB that our test CPU Pentium 4 3.2 MHz sported.

That should give you an idea of how dated the 478 socket is. Intel has already stopped making processors for socket 478, and any stock you can find is leftover stuff.

On the memory support front, we found that the Asus P5P800-MX, the AOpen i915Gm-I, the Gigabyte GA-8I915GV and the GA-8IG1000, as well as the boards that featured the ATi RS350 and RS400 chipset, supported a maximum of 4 GB of DDR 400 MHz RAM.

The Mercury PI915GVM based on the Intel 915GV chipset was the only one to offer two DDR2 memory slots, supporting a maximum of up to 2 GB of 533 MHz RAM, in addition to two DDR slots.

The Intel 845-based boards were FSB limited to 266 MHz, and hence supported up to 2 GB of 266 MHz DDR memory.

Most of the boards supported the dual-channel memory configuration. This makes a difference in memory-intensive applications such as photo-editing applications, and dramatically reduces loading times due to the high available bandwidth.

A majority of the boards had AGP 8x slots, and quite a few of them also had PCIe x16 slots. The

thing to note is that while AGP 8x slots provide an option to upgrade the graphics card and hence, is always preferable over onboard graphics, a PCIe x16 graphics slot is preferable since AGP is slowly but surely making way for PCI-Express and, as you may be aware, PCI-Express graphics cards are already available in the market and will offer a better performance than their AGP counterparts once games utilising their complete bandwidth and capabilities are launched.

Some motherboards such as the Gigabyte GA-8I915GV had only onboard graphics. It had neither an AGP slot nor a PCIe slot and provided no graphics upgrade facility. This is not a good thing, and such motherboards should be avoided unless you are very sure you will not want to go in for better graphics later.

In contrast, motherboards such as the Foxconn 915PL7AE-8S boasted of a PCIe x16 slot in addition to a F.G.E. 8x (Foxconn Graphics Extension) slot, which is compatible with the AGP 4x/8x specifications and lets you plug in a standard AGP card. Thus you are free to choose between an AGP or PCIe graphics card as it suits you. Or, if you are upgrading to this board, you can plug in your old AGP card, and you won't have to invest in a new card. However, this board did not come with onboard graphics.

One motherboard, the ASRock 775Dual-915GL, provided an A.G.I. 8x interface that is compatible with the AGP 8x interface in addition to the A.G.I. Express interface, which is compatible with PCIe x4. A DirectX 9.0c-compatible onboard Intel Extreme Graphics 2 is also present, which is quite decent in terms of performance compared to the older Intel Extreme Graphics.

The speciality of this board is that it lets you plug in both an

AGP as well as a PCIe card at the same time, thus facilitating a multi-monitor working environment. This is just an additional feature which might not be of much value today, since multi-monitor display cards can do the same job at the cost of a single display card.

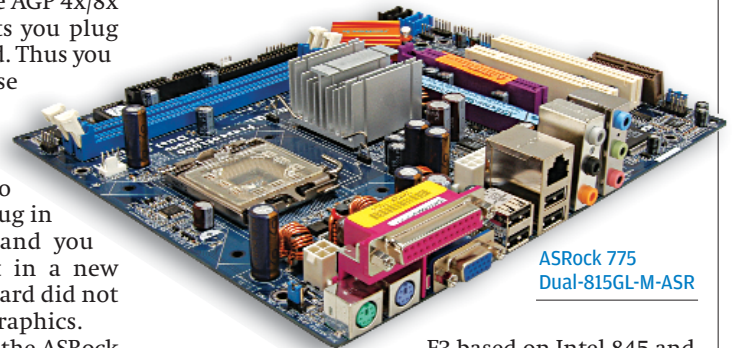
Of the onboard display cards, the ones from ATi, especially the RS400 from HIS, really shone bright because they provided not only commendable performance for onboard graphics, their image quality was also quite good.

On the audio front, none of the motherboards came with minimum of six-channel audio.

The ASRock 775Dual-915GL, Foxconn 915PL7AE-8S, Gigabyte GA-8IG1000 and GA-8I915GV came with as many as 7.1 channels. What this means is that these sound systems are enough to satisfy your most demanding needs, be it in gaming or watching DVDs. With such an onboard system available, one would not find it necessary to invest in a separate sound card.

SATA, which has so far been considered high-end, has finally made its entry into the budget segment. All the boards sported SATA ports, with half the lot having four and the remaining having two ports.

The Gigabyte 8TRS 350MT even supported RAID. Two motherboards, the HIS M48-

ASRock 775
Dual-915GL-M-ASR

F3 based on Intel 845 and the HIS FA61 based on Via P4M800 did not have SATA ports. From a future upgrade perspective, having a SATA port is necessary, since the slower and smaller IDE drives will soon become a thing of the past.

Only the MSI RS350M-ILSR had a FireWire port. A FireWire port is a good thing to have especially if you are into video capturing, because many video